

Thermodynamic instability of siRNA duplex is a prerequisite for dependable prediction of siRNA activities

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ABSTRACT

We developed a simple algorithm, *i-Score* (inhibitory-Score), to predict active siRNAs by applying a linear regression model to 2431 siRNAs. Our algorithm is exclusively comprised of nucleotide (nt) preferences at each position, and no other parameters are taken into account. Using a validation dataset comprised of 419 siRNAs, we found that the prediction accuracy of *i-Score* is as good as those of *s-Biopredsi*, *ThermoComposition21* and *DSIR*, which employ a neural network model or more parameters in a linear regression model. *Reynolds* and *Kato* also predict active siRNAs efficiently, but the numbers of siRNAs predicted to be active are less than one-eighth of that of *i-Score*. We additionally found that exclusion of thermostable siRNAs, whose whole stacking energy (ΔG) is less than -34.6 kcal/mol, improves the prediction accuracy in *i-Score*, *s-Biopredsi*, *ThermoComposition21* and *DSIR*. We also developed a universal target vector, pSELL, with which we can assay an siRNA activity of any sequence in either the sense or antisense direction. We assayed 86 siRNAs in HEK293 cells using pSELL, and validated applicability of *i-Score* and the whole ΔG value in designing siRNAs.

INTRODUCTION

When we study the molecule of our interest, we up- and down-regulate its expression either in cells or in bodies,

and analyze their effects by morphological, physiological and biochemical modalities. Recently, RNA interference (RNAi) has emerged as a simple and robust method to specifically silence a gene expression (1–3). In mammals, 21- to 27-nucleotide (nt) double-stranded RNA or small interfering RNA (siRNA), which is specific to a gene of our interest, is introduced into cells to induce RNAi (4,5).

To achieve efficient and specific gene silencing by siRNA in mammals, an accurate siRNA-designing algorithm is crucial. Numerous algorithms have been reported to date. The algorithm can be arbitrarily divided into two categories: the first-generation algorithms that are based on a small number of observations and the second-generation algorithms that arise from a large number of observations. The first-generation algorithms exploit a variety of siRNA features such as the thermodynamic stability (6,7), base preferences at specific positions (8–12), mRNA secondary structures (13–16) and uniqueness of the target site (17,18). These siRNA features are also summarized in review articles (19–21). The first-generation algorithms disclosed the fundamental requirements for designing active siRNAs.

The prediction accuracies of the first-generation algorithms, however, were not high enough to our satisfaction (22). To improve the prediction accuracy, Huesken and colleagues (23) developed a new algorithm, *Biopredsi*, by applying an artificial neural network model to 2431 siRNAs. *Biopredsi* achieved a high correlation coefficient of 0.66 between the observed and predicted siRNA activities. The artificial neural network modeling, on which *Biopredsi* depends, however, is ‘black box’ in itself, and there is no sense in making further inspections of each parameter.

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In the past two years, the second-generation algorithms emerged by analyzing the Huesken's dataset (24–29). Most algorithms, however, employ complicated mathematical models and depend on calculations that cannot be readily traced, which also prevent us from evaluating these algorithms. Matveeva and colleagues (29) recently compared nine siRNA-designing tools, and concluded that *Biopredsi*, as well as *ThermoComposition* by Shabalina *et al.* (24) and *DSIR* by Vert *et al.* (27), are the best predictors of active siRNAs. The *ThermoComposition* and *DSIR* algorithms employ a linear regression model, which directly indicates nucleotide preferences at each position.

We developed here a simple siRNA prediction algorithm, *i-Score* (inhibitory-Score), based on a linear regression model. The *i-Score* algorithm can predict active siRNAs to the similar extents as *Biopredsi*, *ThermoComposition* and *DSIR*, and is better than the six first-generation algorithms. The *i-Score* algorithm is exclusively composed of the nucleotide preference scores, and is more straightforward than any of the second-generation algorithms. We also found that the whole ΔG value, which represents the stability of the siRNA duplex, ensures accurate prediction of siRNA activities in the four second-generation algorithms, and improves a correlation coefficient to more than 0.7. Additionally, we developed a new validation vector, pSELL, to assay an siRNA activity of any sequence in either the sense or antisense direction simply by synthesizing a pair of oligonucleotides. The synthesized oligonucleotides are inserted into both the pSELL validation vector and the pDual effector vector (30). We validated the efficacy of *i-Score*, as well as the thermostability threshold, by analyzing 86 siRNAs in HEK293 cells.

MATERIALS AND METHODS

Datasets

Dataset A is comprised of 2431 siRNAs reported by Huesken and colleagues (23). The quality of this dataset is ensured by the Gaussian distribution of their potencies. Dataset A is randomly divided into 1600 and 831 siRNAs (Supplementary Table 3). We made five pairs of subsets from dataset A. After confirming that all five pairs gave rise to similar results, we chose a pair of subsets A1600 and A831 without any bias. Dataset B is comprised of 419 siRNAs reported in five other articles (6,8,9,31,32). Each report shows a small number of siRNAs, and the quality of their datasets is variable from report to report.

Training and validation of prediction algorithms

For both modeling and validation analyses, we employed the 'standard least square fitting' functionality of the JMP-IN statistical software Ver. 5.1.1 (SAS Institute, Cary, NC) with its default settings. *i-Score* and *i-Score1600* are trained using dataset A and subset A1600, respectively. As the actual parameters of *Biopredsi* are not published anywhere, we developed a similar scoring system, *s-Biopredsi* for simulated *Biopredsi*, by applying

a single-node neural network model on 2182 siRNAs in subset A (Supplementary Table 3), which is identical to those employed to develop *Biopredsi* (23). We again employed the 'neural network modeling' functionality of the JMP-IN software. We found a high correlation coefficient ($R = 1.0000$) between *Biopredsi* and *s-Biopredsi* with the remaining 249 siRNAs in dataset A. We also obtained a total of 400 *Biopredsi* scores from two independent genes with the *Biopredsi* web server (<http://www.biopredsi.org/>), and compared them to our *s-Biopredsi* scores. Correlation coefficients between *Biopredsi* and *s-Biopredsi* were 0.9999 and 0.9995 for these two genes. Hence *s-Biopredsi* is similar to *Biopredsi*. *s-Biopredsi1600* also employs a single-node neural network modeling, but is trained using subset A1600, so that subset A831 can be used for the validation analysis.

For the receiver operating characteristic (ROC) analysis, we employed the 'logistic regression' functionality of the JMP-IN software, as well as the 'ROC curve' functionality of the SPSS 15.0.1 software (SAS Institute, Cary, NC).

We developed *i-Score* designer (Supplementary Data) with Excel VBA on Windows. We implemented 11 algorithms including *i-Score* by simulating published algorithms and parameters, and confirmed that the *i-Score* designer gives the same scores as those reported by each article. *s-Biopredsi* is similar to, but different from, *Biopredsi* as indicated above. For *ThermoComposition19* and *ThermoComposition21*, the *i-Score* designer calls executable files that have been developed by Matveeva and colleagues (29).

Construction of pSELL and pDual vectors

To validate the prediction accuracy of *i-Score*, we developed a universal target vector, pSELL, which can accommodate any target sequences (Figure 5A). We first excised EGFP from pIRES2-EGFP (Clontech, Mountain View, CA, USA) and placed it upstream of IRES. We then inserted the firefly luciferase gene downstream of IRES, and made pCMV-EGFP-IRES-Luc. We additionally inserted a part of the LacZ gene between the target-cloning site and IRES so that IRES-binding proteins do not mask the upstream target sequence. We next substituted the SV40 promoter for the CMV promoter, and made pSV40-EGFP-LacZ-IRES-Luc (pSELL). We mutated an extra HindIII restriction site within IRES, and exploited the BglII, HindIII and BamHI restriction sites between EGFP and IRES to accommodate a target sequence in either the sense or antisense direction.

We also constructed an siRNA-generating vector, pDual, according to Zheng and colleagues (30). Briefly, we inserted the mouse U6 promoter and the human H1 promoter in opposite directions in pBluescript SK(-) (Stratagene, La Jolla, CA, USA). We also introduced a BglII site so that the siRNA sequence can be inserted between the HindIII and BglII sites flanked by the U6 and H1 promoters.

Cell culture and transfection

Human embryonic kidney (HEK) 293 cells were maintained in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bovine serum. In all assays, 1.5×10^5 HEK293 cells were plated on 24-well dishes 12 h before transfection. Cells were subsequently transfected with 0.3 μ g of the pDual effector vector, 0.03 μ g of the pSELL target vector and 0.03 μ g of phRL-TK encoding the Renilla luciferase (Promega, Madison, WI, USA) using the FuGENE6 Transfection Reagent (Roche Applied Science, Basel, Switzerland). After incubation for 3 days, firefly and Renilla luciferase activities were measured using the Dual-Luciferase Reporter Assay System (Promega). Silencing activity (% inhibition) of each siRNA is calculated by dividing the relative luciferase activity in the presence of the pDual target vector by the relative luciferase activity in the presence of a control pDual vector lacking an insert.

RESULTS

Development of a simple algorithm, *i-Score*, for prediction of siRNA activities

In order to develop a simple algorithm to predict active siRNAs, we collated two datasets (Supplementary Table 3). We applied a linear regression model to dataset A comprised of 2431 siRNAs to construct a prediction algorithm, *i-Score* for inhibitory score, and validated it with dataset B comprised of 419 siRNAs. Although dataset A is comprised of 21-nt siRNA sequences, we eliminated 2-nt overhangs at the 3' end of the antisense strand and employed 19 nt that make an siRNA duplex, in order to validate our algorithm with dataset B, which is comprised of 19-nt sequences. Our preliminary analysis using subsets A1600 and A831 demonstrated that 19-nt analysis is as good as 21-nt analysis in our linear regression model (data not shown).

Our linear regression model determines nucleotide preferences at each position of siRNA, which is then used as scoring parameters to calculate *i-Score* (Figure 1A). We normalized the scoring parameters to give the best and worst *i-Scores* of 100 and 0, respectively (Figure 1B and Supplementary Table 1). The scoring parameters directly demonstrate which nucleotides are preferred at which positions. Previous reports address the importance of G/C at positions 18 and 19 on the antisense strand, as well as a stretch of A/T at the 5' end on the antisense strand (6,7). Our results also conform to this notion. In addition, highly positive and negative scoring parameters in our analysis are located at previously reported preferred and unfavorable nucleotides, respectively (6,23,24,27).

Comparison of *i-Score* with three other second-generation algorithms of *s-Biopredsi*, *Thermocomposition21* and *DSIR*

To test how efficiently *i-Score* predicts siRNA activities, we plotted 419 siRNA activities in dataset B against *i-Scores* (Figure 2A). We similarly plotted the observed

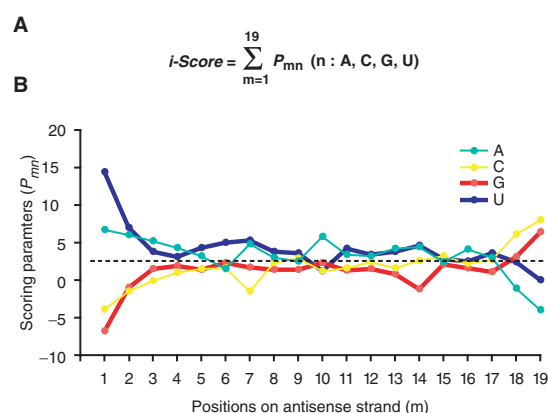


Figure 1. Definition of *i-Score* (A) and scoring parameters at each position (B). Parameters are normalized to give the best and the worst *i-Scores* of 100 and 0, respectively. The average score at each position is 2.60 (dotted line). Scores above 2.60 indicate preferred nucleotides. '*P*' is a probability score of nucleotide '*n*' at position '*m*' on the antisense strand (Supplementary Table 1).

and predicted siRNA activities with *s-Biopredsi*, *Thermocomposition21* and *DSIR* (Figure 2B–D). None of these algorithms employs dataset B in the process of training parameters. The correlation coefficients indicate that *i-Score* is as good as the three other second-generation algorithms (Table 1). For all the algorithms, the correlation coefficients with dataset A are superior to those with dataset B, implying possible overfitting with the training dataset A.

The ROC curve is a plot of sensitivity versus 1-specificity, and is widely applied to compare efficiencies of different algorithms in a variety of biomedical fields. The ROC analysis with dataset B also demonstrates that the prediction accuracies of the four algorithms are not statistically different (Figure 2E).

We also examined the correlation coefficients among *i-Score*, *s-Biopredsi*, *ThermoComposition21* and *DSIR*, and found that *i-Score*, *s-Biopredsi* and *DSIR* are close to each other, whereas *ThermoComposition21* is unique compared to the other three algorithms (Supplementary Table 2).

Comparison of the first- and second-generation algorithms

We next compared *i-Score* with *s-Biopredsi*, as well as with six other first-generation algorithms of *Reynolds*, *Ui-Tei*, *Amarzguioui*, *Katoh*, *Hsieh* and *Takasaki*, using subset A831 (Supplementary Figure 1 and Table 2). Both *i-Score* and *s-Biopredsi* are classified into the second-generation algorithms, because these are based on dataset A. As subset A831 is included in the training dataset A for *i-Score* and *s-Biopredsi*, we employed *i-Score1600* and *s-Biopredsi1600*, which we trained with subset A1600, in order to strictly avoid overfitting.

When we sort siRNAs in descending order of predicted scores and choose siRNAs above a given threshold, the ratio of active siRNAs among the selected siRNAs is inversely correlated with the number of selected siRNAs for all the algorithms (Table 2). Namely, if an algorithm chooses more siRNAs, the chance of obtaining active

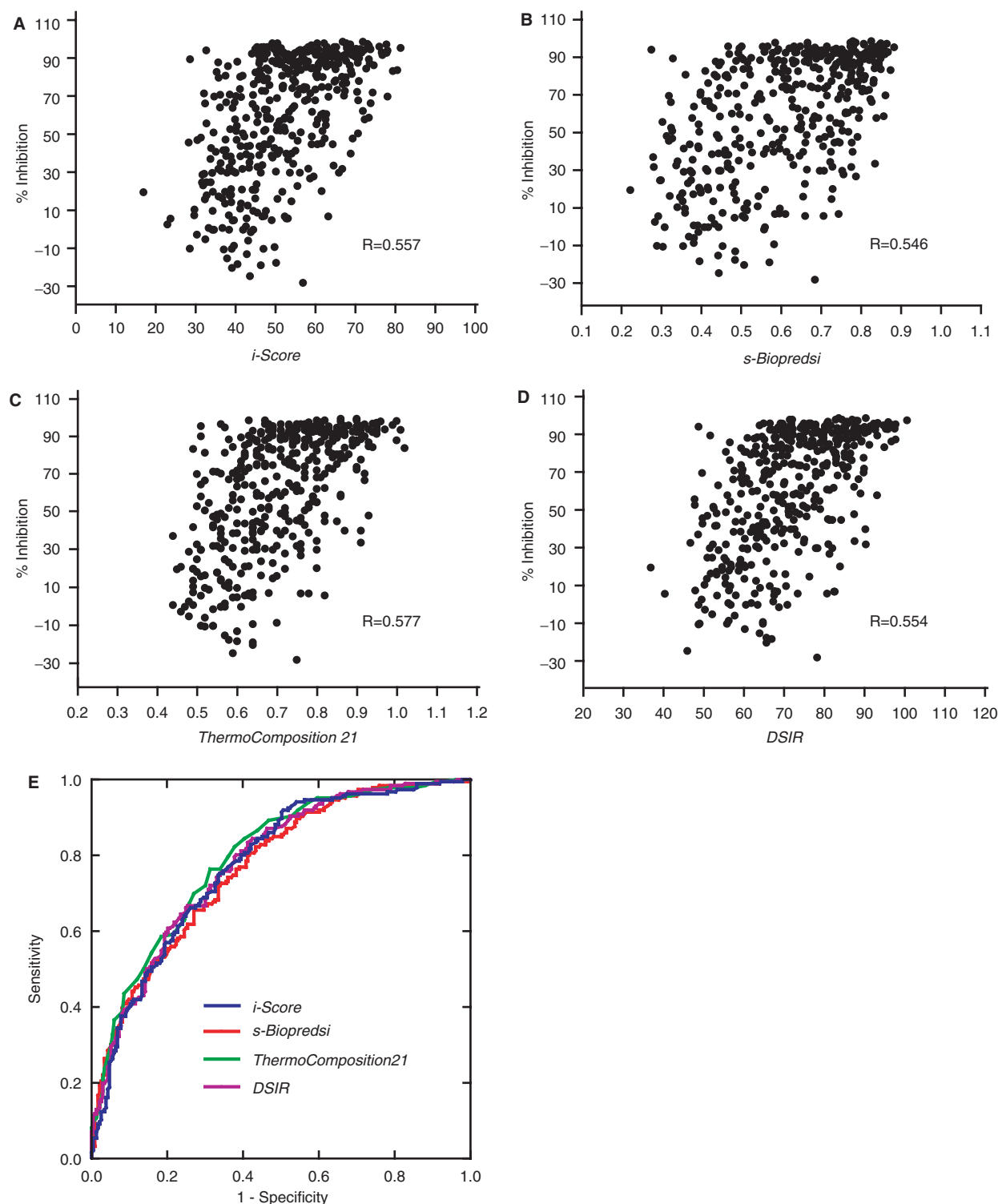


Figure 2. Observed siRNA activities in dataset B are plotted against predicted siRNA activities by *i*-Score (A), *s*-Biopredsi (B), ThermoComposition21 (C) and DSIR (D). ‘R’ values represent the Pearson correlation coefficients, which are also indicated in Table 1. (E) ROC curves of the four algorithms. Areas under the curves (AUC) of *i*-Score, *s*-Biopredsi, ThermoComposition21 and DSIR are 0.776 (95% confidence interval, 0.732–0.820), 0.770 (0.726–0.814), 0.795 (0.753–0.837) and 0.781 (0.738–0.825), respectively. There are no statistical differences of AUCs among the four algorithms.

siRNAs becomes less. siRNAs with *i*-Score₁₆₀₀ ≥ 65.9 constitute 8.8% of subset A831, and 90% of the siRNAs in this category are active. Similarly, siRNAs with *s*-Biopredsi₁₆₀₀ ≥ 0.807 comprise 8.8% of subset A831,

and 90% are active. In the current analysis, Reynolds and Katoh reach a ~90% success rate, whereas Ui-Tei, Amarzguoui, Hsieh and Takasaki do not (Table 2). siRNAs with Reynolds ≥ 9 constitute 1.1% of subset

Table 1. Pearson correlation coefficients between observed and predicted siRNA activities by four second-generation algorithms

	Dataset A	Dataset B
<i>i-Score</i>	0.635	0.557
<i>s-Biopredsi</i>	0.665	0.546
<i>ThermoComposition21</i>	0.635	0.577
<i>DSIR</i>	0.687	0.554

Dataset A is used to train all the algorithms, whereas dataset B is not used as a training dataset in any algorithms.

A831, and 88.9% are active. Similarly, siRNAs with *Katoh* > 101.1 comprise 0.7% of subset A831, and 90% are active. Therefore, when we design siRNAs expecting a ~90% success rate, *i-Score1600* and *s-Biopredsi1600* would be able to predict at least eight times more numbers of siRNAs compared to *Reynolds* and *Katoh*. This likely represents the most beneficial advantage of the second-generation algorithms over the first-generation ones.

Whole ΔG value, as an indicator of thermostability of siRNA duplex, is a key determinant of accurate prediction

We next sought for another parameter that potentially improves the correlation coefficient between the observed and predicted siRNAs with *i-Score*. In the scattered plot of the observed and predicted siRNA activities, we observe a subgroup of effective siRNAs in which *i-Scores* are falsely low (area L in Figure 3A and B). As siRNAs in area L make the correlation coefficient low, we searched for a shared feature among siRNAs in area L. For this purpose, we calculated 12 parameters for each siRNA: stacking energy (ΔG) of the secondary structure of siRNA (38), maximum GC stretch within siRNA, %GC contents spanning the antisense positions 1–19, 3–7 and 1–17, and stacking energies (ΔG) spanning the antisense positions 1–19, 3–17, 11–19, 1–9, 11–17, 3–9, 1–5 and 1–17. We divided the 419 siRNAs in dataset B into two subsets by gradually changing the threshold for each parameter, and analyzed the correlation coefficients between the two subsets.

This analysis revealed that siRNAs with a stable stacking energy at positions 1–19 (the whole ΔG value) tend to give rise to a low correlation coefficient. A contour plot of the whole ΔG values also illustrates that siRNAs in area L have stable stacking energies of ≤ -37 kcal/mol (Figure 3A and B). The contour plot further demonstrates that thermodynamically unstable siRNAs with the whole ΔG values > -31 kcal/mol tend to stay close to the linear regression line, which also points to the notion that exclusion of thermostable siRNAs makes the correlation coefficient high. Indeed, the correlation coefficient goes higher, when we gradually exclude thermostable siRNAs by elevating the threshold of the whole ΔG values from -52.0 up to -34.6 kcal/mol for *i-Score*, *s-Biopredsi*, *ThermoComposition21* and *DSIR* (Figure 3C). The correlation coefficient, however, goes down when the whole ΔG threshold goes further up, likely because only a limited number of siRNAs can be included in the analysis. Although the whole ΔG values are well correlated

Table 2. Comparison of eight algorithms using subset A831

Threshold	% of effective siRNAs ^a	No (%) of siRNAs matching the threshold	Pearson correlation coefficient
<i>i-Score1600</i> ^{b,c}			0.592
≥ 65.9	90	72 (8.8%)	
≥ 63.0	80	117 (14.1%)	
≥ 59.4	75	166 (20.0%)	
<i>s-Biopredsi1600</i> ^{b,c}			0.618
≥ 0.807	90	73 (8.8%)	
≥ 0.767	80	141 (17.0%)	
≥ 0.734	75	197 (23.7%)	
<i>Reynolds</i> ^d			N.D.
≥ 9	88.9	9 (1.1%)	
≥ 8	80.4	46 (5.5%)	
≥ 7	71.0	107 (12.9%)	
<i>Ui-Tet</i> ^d			N.D.
Ia	73.4	64 (7.7%)	
Ia + Ib	68.0	125 (15.0%)	
<i>Amarzguoui</i> ^d			N.D.
≥ 5	81.0	21 (2.5%)	
≥ 4	61.3	80 (9.6%)	
≥ 3	64.8	179 (21.5%)	
<i>Katoh</i> ^c			0.427
≥ 101.1	90	6 (0.7%)	
≥ 87.5	80	31 (3.7%)	
≥ 79.5	75	80 (11.1%)	
<i>Hsieh</i>			N.D.
$= 4$	50.0	2 (0.2%)	
≥ 3	51.2	41 (4.9%)	
≥ 2	52.7	184 (22.1%)	
<i>Takasaki</i>			0.174
n.a. ^c	90	n.a. ^c	
n.a. ^c	80	n.a. ^c	
≥ 17.1	75	4 (0.5%)	

Note that *i-Score1600*, *s-Biopredsi1600*, *Reynolds* and *Katoh* predict active siRNAs with ~90% accuracy, but the chances of predicting such active siRNAs in a given mRNA with *Reynolds* and *Katoh* are ~1/8 and ~1/12, respectively, of those with *i-Score1600* and *s-Biopredsi1600*. *ThermoComposition21* and *DSIR* are not included in this analysis, because we cannot avoid overfitting for these two algorithms for subset A831.

^aRatios of experimentally proved active siRNAs among siRNAs predicted to be active according to variable thresholds of eight different algorithms. The experimentally proved active siRNAs are arbitrarily defined to those suppressing the gene expression levels to less than 25% of a control.

^bThe *i-Score1600* and *s-Biopredsi1600* scores are different from *i-Score* and *s-Biopredsi*. In order to avoid overfitting for these two algorithms, the modeling algorithms of *i-Score* and *s-Biopredsi* are applied to subset A1600 to calculate scoring parameters for *i-Score1600* and *s-Biopredsi1600*, respectively. The Pearson correlation coefficients between *i-Score* and *i-Score1600*, and *s-Biopredsi* and *s-Biopredsi1600* are 0.990 and 0.985, respectively.

^cAs *i-Score1600*, *s-Biopredsi1600*, *Katoh* and *Takasaki* are continuous numeric scores, thresholds are arbitrarily set so that 90, 80 and 75% siRNAs above the threshold are experimentally proved active. For example, for *i-Score1600*, all the 831 siRNAs in subset A831 are sorted in descending order of *i-Score1600*. The ratio of experimentally proved active siRNAs decreases with decreasing *i-Score1600*. When the lower limit of *i-Score1600* is set to 65.9, 72 siRNAs are included in this category and 65 suppress the gene expression levels to less than 25% of a control. This is how we set an *i-Score1600* threshold for 90% (65/72). The 72 siRNAs comprise 8.8% of the 831 siRNAs in subset A831. This indicates that when we synthesize a siRNA with *i-Score1600* of 65.9 or higher, we can expect that the chance of obtaining an active siRNA is 90%.

^dFor algorithms yielding ordinal or nominal numbers, the indicated ranks are used as the thresholds. For example, for *Reynolds*, the number of siRNAs equal to or higher than the score of 9 is 9, which comprises 1.1% of the 831 siRNAs, and 8 (88.9%) of the 9 siRNAs suppress gene expression levels to less than 25% of a control.

^eNo subgroup of siRNAs matches the criteria of $\geq 90\%$ or $\geq 80\%$ prediction accuracy.

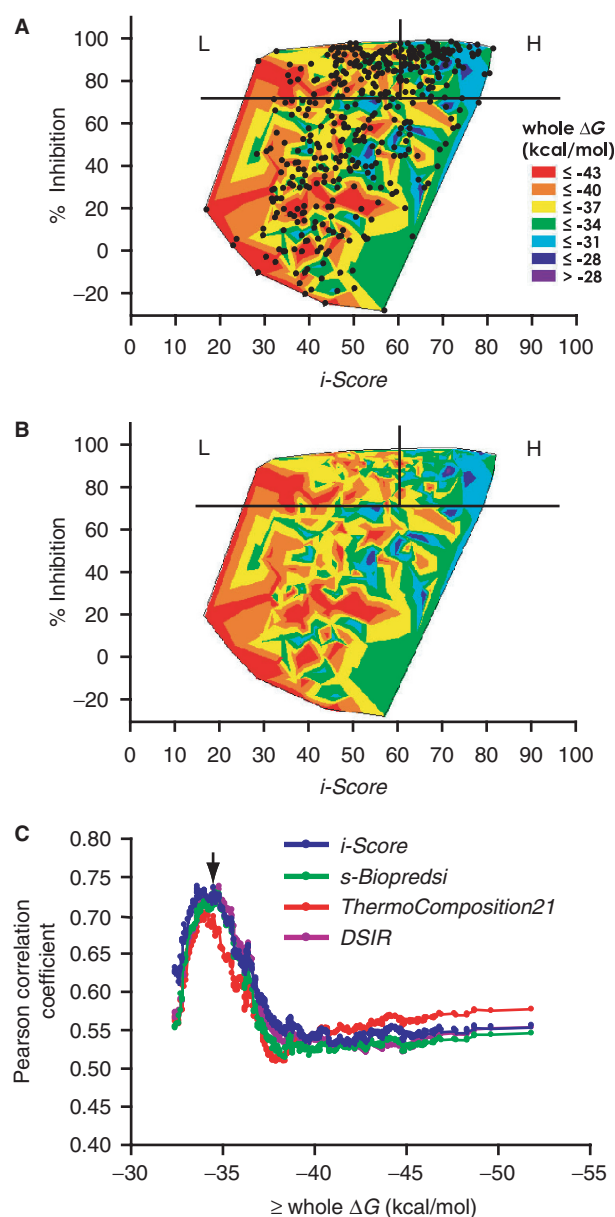


Figure 3. Contours of the whole ΔG values overlaid on the scattered plots of observed and predicted siRNA activities with (A) or without (B) dots indicating individual siRNAs for dataset B. Area H indicates siRNAs for which *i-Score* successfully predicts siRNA activities, whereas area L indicates siRNAs for which *i-Score* falsely predicts inactive siRNAs. Thermodynamically unstable siRNAs with high whole ΔG values tend to align on a linear regression line (data not shown) going from the bottom left up to the top right, and form a cluster of well-predicted siRNAs (area H) at the top right corner. On the other hand, thermostable siRNAs stay away from the regression line, and form a cluster of poorly predicted siRNAs (area L) at the top left corner. The plots are drawn by the 'contour plot' functionality of the JMP-IN software. (C) Correlation coefficients between the observed and predicted siRNA activities in subgroups of siRNAs whose whole ΔG values are equal to or more than the indicated values. For example, there are 101 siRNAs whose whole ΔG values are equal to or more than -34.6 kcal/mol. The correlation coefficient of the 101 siRNAs with *i-Score* is 0.723, and hence 0.723 is plotted on -34.6 (arrow). This is also illustrated in Figure 4. The correlation coefficient tends to go down when the whole ΔG threshold further goes higher, likely because we can include a limited number of siRNAs in the analysis.

with %GC contents ($R = 0.98$), the whole ΔG value is a better discriminator than the %GC content (data not shown).

We found that siRNAs with the whole ΔG value of -34.6 kcal/mol and higher result in a correlation coefficient of 0.723 (Figure 4A), whereas the remaining thermostable siRNAs give rise to a correlation coefficient of 0.514 for dataset B (Figure 4B). ROC analysis also demonstrates that a data subset comprised of unstable siRNAs (≥ -34.6 kcal/mol) gives rise to a markedly higher AUC than that of stable siRNAs (Figure 4C). These analyses all point to the notion that exclusion of thermostable siRNAs improves the correlation between the observed and predicted siRNA activities in all the algorithms.

Genome-wide prediction of active siRNAs with *i-Score*

As shown in Table 2, 90% of siRNAs are active, if we choose siRNAs with *i-Score* ≥ 65.9 for subset A831. Additionally, as shown in Figure 4, the whole ΔG value of -34.0 kcal/mol differentiates between successfully and falsely predicted siRNAs. We can thus expect that more than 90% of siRNAs are active, if we set the thresholds of *i-Score* > 66 and the whole ΔG value > -34.0 kcal/mol.

In order to test if *i-Score* can indeed predict active siRNAs in human genes even after we impose a threshold for the whole ΔG value, we applied variable thresholds of *i-Score* and the whole ΔG value to all the transcripts in the NCBI RefSeq Database Build 35.1 (Table 3). When we choose siRNAs with *i-Score* > 65 and the whole ΔG value > -34.0 kcal/mol, we expect to predict 65 active siRNAs per mRNA. Under these conditions, 89.2% of human transcripts are expected to have one or more active siRNAs. These results suggest that *i-Score* potentially predicts active siRNAs in most human genes.

i-Score designer

We developed an Excel VBA program, the *i-Score* designer (Supplementary Data), which calculates 11 different siRNA-designing scores including *i-Score* for all possible siRNA sequences within a gene of our interest or for individually entered siRNA sequences. The program also calculates the whole ΔG value and five other parameters.

A new validation method for siRNA activity

To validate the prediction accuracy of *i-Score*, we developed a universal target vector, pSELL, which accommodates any target sequence in either the sense or antisense direction (Figure 5A). We also constructed the pDual effector vector (30). With pSELL and pDual, a single pair of synthesized oligonucleotides can be inserted into both the target and effector vectors. We constructed 86 pairs of pSELL and pDual vectors (Supplementary Table 4), and assayed their effects in HEK293 cells by measuring the luciferase activities. As expected, *i-Score* predicted siRNA activities more accurately for those with thermodynamically unstable siRNAs (Figure 5B) than those with thermostable siRNAs (Figure 5C).

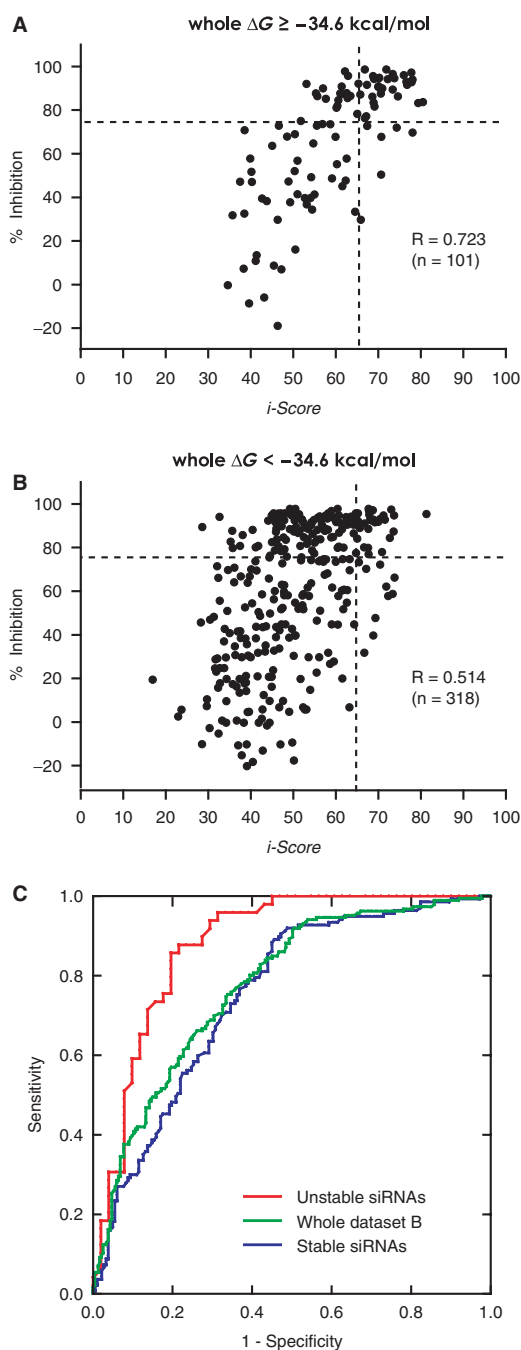


Figure 4. Scattered plots of observed and predicted siRNAs categorized by the whole ΔG values for datasets B. The threshold of whole ΔG values is indicated on top of each panel. i -Score predicts activities of thermodynamically unstable siRNAs (A) more accurately than those of thermostable siRNAs (B). (C) ROC curves of thermodynamically unstable (≥ -34.6 kcal/mol) and stable (< -34.6 kcal/mol) siRNAs in dataset B using i -Score. AUCs of unstable and stable siRNAs are 0.882 (95% confidence interval, 0.814–0.950) and 0.750 (0.697–0.803), respectively. AUC of the whole dataset B is 0.776 (0.732–0.820).

DISCUSSION

A simple algorithm, i -Score, to predict active siRNAs

We developed a simple siRNA-designing algorithm, i -Score, by applying a linear regression model to 2431

siRNAs. The correlation coefficient that we achieved with dataset B was as high as those of *s-Biopredsi* (23), *ThermoComposition21* (24,29) and *DSIR* (27). Comparison of i -Score with *s-Biopredsi*, as well as with *Reynolds* (8), *Ui-Tei* (9), *Amarzguioui* (11), *Katoh* (33), *Hsieh* (12) and *Takasaki* (10), using subset A831 demonstrated that i -Score, *s-Biopredsi*, *Reynolds* and *Katoh* can readily predict active siRNAs with $\sim 90\%$ accuracy. Additionally, both i -Score and *s-Biopredsi* predict at least eight times more numbers of active siRNAs than *Reynolds* and *Katoh*. We analyzed the prediction efficiencies under conditions where no overfitting is allowed.

The advantage of i -Score over the others is that i -Score only takes into account the nucleotide preferences at each position and employs no other parameters, which makes the calculation of i -Score simple and easy to trace. In addition, we can visually inspect which nucleotides are better than the others at a specific position. Teramoto and colleagues (34) report that short motifs of 1–3 nt without positional information provide enough parameters for designing siRNAs. Vert and colleagues (27) also report that inclusion of short motifs in addition to the position-specific nucleotide preferences improves the prediction accuracy, and made the *DSIR* scores using a subset of our dataset A. *DSIR* is indeed superior to i -Score for dataset A according to our analysis (Table 1). With dataset B, however, prediction accuracy of *DSIR* is not as good as that of i -Score (Table 1). This may represent overfittings of *DSIR* with dataset A. Otherwise, either i -Score or *DSIR* is applicable to specific datasets but not to the others, but the underlying causes remain elusive.

Exclusion of thermostable siRNAs improves the prediction accuracy

Ladunga (28) reports that the whole ΔG values are correlated with siRNA activities. We also observe a correlation coefficient of $R = 0.279$ between the whole ΔG values and the siRNA activities for dataset A. Inclusion of the whole ΔG value as an independent parameter in our linear regression model, however, fails to improve correlation coefficients for our validation datasets (data not shown). This is likely because the whole ΔG value is already represented in i -Score, and indeed the correlation coefficient between these two parameters is 0.445.

We found that exclusion of thermostable siRNAs is beneficial in i -Score, *s-Biopredsi*, *ThermoComposition21* and *DSIR* (Figure 3C). We also found that even after we impose a threshold of the whole ΔG values, we can predict enough numbers of active siRNAs for all the human transcripts with i -Score (Table 3). Our analysis demonstrates that the whole ΔG value rather serve as a determinant for successful prediction of siRNA activities.

In addition, our analysis demonstrates that some thermostable siRNAs are indeed active (area L in Figure 3A and B), and that no current algorithm can accurately predict their activities. This likely represents the presence of another cluster of siRNAs that potentially

Table 3. Genome-wide prediction of active siRNAs by *i-Score*

<i>i-Score</i>	Whole ΔG	siRNAs/kb ^a	siRNAs/mRNA ^b	Best <i>i-Score</i> ^c	10th <i>i-Score</i> ^d
>65	– ^e	82.9	132 (99.8%)	82.6 ± 4.3	75.6 ± 4.7
>65	>–34	50.5	65 (95.9%)	82.2 ± 5.3	74.2 ± 7.2
>65	>–30	26.7	33 (89.2%)	81.8 ± 5.9	72.8 ± 7.8
>70	– ^e	34.0	53 (98.8%)	– ^f	– ^f
>70	>–34	24.2	33 (94.0%)	– ^f	– ^f
>70	>–30	14.1	19 (86.9%)	– ^f	– ^f

^aAverage number of active siRNAs per kb of the human RefSeq mRNAs. The NCBI RefSeq Database Build 35.1 includes 40 768 mRNAs, and we analyzed each alternatively spliced transcript as an independent mRNA. The total number of nucleotides that we analyzed is 100 738 984, and the mean and SD of the mRNA length is 2471 ± 2039 bases.

^bMedian number of active siRNAs per mRNA in the human RefSeq database. As the numbers of active siRNAs do not follow a Gaussian distribution, median numbers are represented. The number in parenthesis indicates a percentage of mRNAs among the 40 768 transcripts, harboring at least one siRNA meeting the indicated criteria.

^cMean and SD of the highest *i-Score* for each mRNA. mRNAs with no siRNAs above the indicated whole ΔG threshold are excluded from the analysis.

^dMean and SD of the 10th highest *i-Score* for each mRNA. mRNAs with less than 10 siRNAs above the indicated whole ΔG threshold are excluded from the analysis.

^eNo threshold is placed for the whole ΔG .

^fBecause means of the best and the 10th highest *i-Scores* are independent of the *i-Score* threshold, values are indicated only on rows of *i-Score* >65.

requires unidentified parameters to precisely predict their activities. Indeed, Krueger and colleagues (35) recently reported that, in addition to the siRNA sequence and its concentration, unidentified characteristics specific to the target gene are likely to have a significant influence on the siRNA activities.

***i-Score* designer**

i-Score predicts on average 65 active siRNAs per mRNA in the genome-wide analysis (Table 3). Other algorithms would also predict similar numbers of active siRNAs. Among these siRNAs, we usually choose a single algorithm and synthesize one or two siRNAs with the best scores. As no algorithm has achieved ~100% prediction accuracy, we usually wonder if the selected siRNAs are indeed active or not. If all the siRNAs, whose scores of an algorithm of our choice are above a predefined threshold, can be analyzed by the other algorithms, the chance of obtaining active siRNAs would become high. Most of the currently available siRNA-designing programs, however, do not provide scores of all available siRNAs within a given gene. To overcome this problem, we implemented 11 different scores and six parameters in *i-Score* designer (Supplementary Data). The *i-Score* designer readily demonstrates how siRNAs selected by an algorithm of our choice are evaluated by the other algorithms. As far as we know, no siRNA-designing program offers such functionality.

pSELL and pDual as a new validation tool of siRNA activity

To validate the prediction accuracy of siRNA-designing algorithms including *i-Score*, we developed a universal target vector, pSELL, in which the target sequence can be inserted in the sense or antisense direction. A combination of the pSELL target vector and the pDual effector vector (30) is cost-effective, because a single pair of oligonucleotides spanning an siRNA sequence of our interest can be inserted into both vectors. Hung and colleagues (36)

report a similar cost-effective strategy employing pDual. Their target vector carries a reporter gene of either EGFP or luciferase followed by a target sequence in its 3' untranslated region. On the other hand, pSELL carries EGFP, the target sequence IRES and the luciferase gene (Figure 5A). First, with pSELL, we can substitute our gene of interest for EGFP, and can quantify the siRNA activity against the full-length mRNA by measuring the expression level of the target mRNA as well as the luciferase activity. Second, pSELL enables us to measure both EGFP and luciferase activities in a single experiment. Third, a unique feature of pSELL is that it accommodates the target sequence in either the sense or antisense direction, which facilitates analysis of an effect of an siRNA on the antisense strand. As a large proportion of mammalian genes harbor antisense transcripts, which regulate the expression levels of the sense transcripts (37), the ability to assay an effect of an siRNA on the antisense strand will become more and more essential when we knock down a gene of our interest.

SUPPLEMENTARY DATA

Supplementary Data are available at NAR Online.

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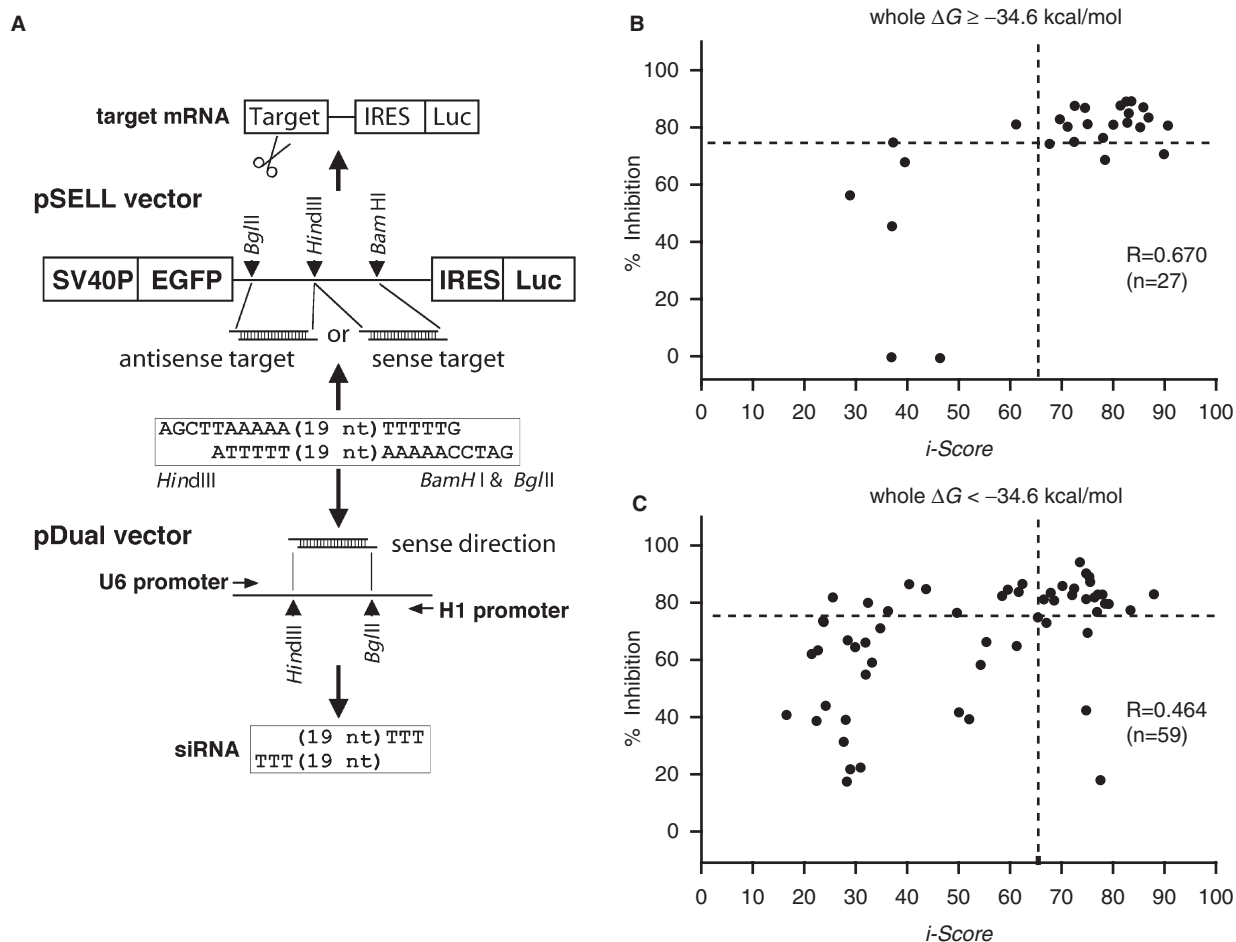


Figure 5. *In vitro* validation of *i*-Score using pSELL and pDual vectors. (A) The pSELL target vector carries the SV40 promoter, the *EGFP* gene, the target-cloning site, IRES and the firefly luciferase gene. Synthesized oligonucleotides should carry HindIII- and BamHI/BglII-competent restriction sites at both ends. pSELL can accommodate the synthesized oligonucleotides in either direction. The pDual effector vector harbors the U1 and H1 promoters in opposite directions, and the cloning site between them (30). pDual accommodates the same oligonucleotides as pSELL in the sense direction. siRNA synthesized by pDual works on the target mRNA synthesized by pSELL. We can thus quantify the siRNA activity by measuring the activity of either EGFP or the firefly luciferase. Scattered plots of 86 observed and predicted siRNA activities above (B) and below (C) the whole ΔG value of -34.6 kcal/mol. The correlation coefficient of thermodynamically unstable siRNAs (B) is better than that of thermostable siRNAs (C).

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